

Reconnaissance Tools

Documentation

This document provides an overview of four widely used reconnaissance tools: Nmap, Subfinder, httpx, and Amass. It includes purpose, use case, input & output, features, installation on Kali Linux, common commands, and a recommended workflow.

Nmap

Overview

Active network scanner used to discover hosts, open ports, running services, OS detection, and NSE scripts.

Use Case, Input & Output

Use Case	Mapping a target's live hosts and open services during the active scanning phase of a penetration test, to identify entry points before deeper testing.
Input	A target IP address, hostname, domain, or CIDR range, along with optional flags for port ranges, scan type, and NSE scripts.
Output	A report of live hosts, open/closed/filtered ports, detected services and versions, OS guesses, and NSE script results, viewable on screen or saved via -oA in normal, XML, and grepable formats.

Key Features

- Host discovery
- Port scanning
- Service/version detection
- OS detection
- NSE scripting

Installation (Kali Linux)

```
sudo apt update
sudo apt install nmap -y
nmap --version
```

Common Commands

Basic scan:

nmap example.com

Target:

Command:

Hosts		Services		Nmap Output		Ports / Hosts		Topology		Host Details		Scans	
OS	Host	Port	Protocol	State	Service	Version							
	192.168.56.1	135	tcp	open	msrpc	Microsoft Windows RPC							
		139	tcp	open	netbios-ssn	Microsoft Windows netbios-ssn							
		445	tcp	open	microsoft-ds								
		8000	tcp	open	http	Splunkd httpd							
		8089	tcp	open	http	Splunkd httpd							

Service detection:	nmap -sV example.com <table><thead><tr><th colspan="2">Nmap Output</th><th>Ports / Hosts</th><th>Topology</th><th>Host Details</th><th>Scans</th></tr><tr><th>Port</th><th>Protocol</th><th>State</th><th>Service</th><th>Version</th><th></th></tr></thead><tbody><tr><td>80</td><td>tcp</td><td>open</td><td>tcpwrapped</td><td></td><td></td></tr><tr><td>443</td><td>tcp</td><td>open</td><td>tcpwrapped</td><td></td><td></td></tr></tbody></table>	Nmap Output		Ports / Hosts	Topology	Host Details	Scans	Port	Protocol	State	Service	Version		80	tcp	open	tcpwrapped			443	tcp	open	tcpwrapped		
Nmap Output		Ports / Hosts	Topology	Host Details	Scans																				
Port	Protocol	State	Service	Version																					
80	tcp	open	tcpwrapped																						
443	tcp	open	tcpwrapped																						
OS detection:	sudo nmap -O example.com Operating System Name: Fortinet FortiGate-50B or 310B firewall Accuracy: Ports used Port- 443 Protocol- tcp State: open OS Classes Type Vendor OS Family OS Generation Accuracy firewall Fortinet embedded																								
Aggressive scan:	nmap -A example.com																								
All ports:	nmap -p- example.com Scanning 192.168.56.1 [65535 ports] Discovered open port 139/tcp on 192.168.56.1 Discovered open port 135/tcp on 192.168.56.1 Discovered open port 445/tcp on 192.168.56.1 Discovered open port 49668/tcp on 192.168.56.1 Discovered open port 49665/tcp on 192.168.56.1 Discovered open port 5040/tcp on 192.168.56.1 Discovered open port 49667/tcp on 192.168.56.1 Discovered open port 49666/tcp on 192.168.56.1 Discovered open port 61688/tcp on 192.168.56.1 Discovered open port 8191/tcp on 192.168.56.1 Discovered open port 8000/tcp on 192.168.56.1 Discovered open port 49664/tcp on 192.168.56.1 Discovered open port 49706/tcp on 192.168.56.1 Discovered open port 8089/tcp on 192.168.56.1 Discovered open port 7680/tcp on 192.168.56.1 Completed SYN Stealth Scan at 11:38, 4.06s elapsed (65535 total ports) Initiating Service scan at 11:38 Scanning 15 services on 192.168.56.1																								
Save output:	nmap -oA scan example.com																								

Subfinder

Overview

Passive subdomain enumeration tool from ProjectDiscovery.

Use Case, Input & Output

Use Case	Early-stage OSINT reconnaissance to quickly discover a target organization's subdomains without touching their infrastructure directly, expanding the attack surface for further testing.
Input	A single root domain, or a file containing a list of domains; optional API keys for third-party sources configured via the provider config file.
Output	A list of discovered subdomains printed to the console or saved to a file (plain text or JSON), suitable for piping into tools like httpx or Amass.

Key Features

- Passive OSINT
- Fast enumeration
- API support
- JSON output

Installation (Kali Linux)

```
sudo apt update
sudo apt install subfinder -y
subfinder -version
```

Common Commands

Enumerate:

subfinder -d tesla.com

```
(dharmik@kali)~$ subfinder -d tesla.com

projectdiscovery.io

[INF] Current subfinder version v2.14.0 (latest)
[INF] Loading provider config from /home/dharmik/.config/subfinder/provider-config.yaml
[INF] Enumerating subdomains for tesla.com
mx.digitalcontent.tesla.com
bettertime-stage.tesla.com
apmv3-cdn-go.tesla.com
npuapt-prd.usw.vn.cloud.tesla.com
uploads.github-fw.tesla.com
static-assets-contactus-comm-preferences.tesla.com
gateway-public-service-center-eng5a-fleet.eng.vn.cloud.tesla.com
analytics.hcaptcha.tesla.com
livestream.tesla.com
m.location-services-prd.tesla.com
autodiscover.epuca-prd.euw1.vn.cloud.tesla.com
assets.github-it.tesla.com
issue-tracker.tesla.com
akamai-apigateway-vehicleextinfofw-prdsvc-st.tesla.com
go.location-services-prd.tesla.com
hub.epc.tesla.com
webmail.auth.tesla.com
diner-webapp-stage.tesla.com
discovery-service.bottlerocket.tesla.com
kronos-dev.tesla.com
mail.assistant-api.prd.usw2.vn.cloud.tesla.com
ns3.akamai-apigateway-vfx.tesla.com
codeload.github.tesla.com
api-kronos-dev.tesla.com
k8s.hermes-stream-api.prd.eu.vn.cloud.tesla.com
private.epc.tesla.com
live-data.prd.usw2.fn.tesla.com
render.github-ap.tesla.com
```

subfinder -d example.com -o teslasubdomains.txt

Save output:

```
(dharmik@kali)-[~]
$ subfinder -d tesla.com -o teslasubdomains.txt

Places
Tesla
projectdiscovery.io

[INF] Current subfinder version v2.14.0 (latest)
[INF] Loading provider config from /home/dharmik/.config/subfinder/provider-config.yaml
[INF] Enumerating subdomains for tesla.com
epuca-prd.usw2.vn.cloud.tesla.com
reply.github-it.tesla.com
mfs-synergy.tesla.com
gfb.vdi.tesla.com
admin.apigateway-logging.tesla.com
apacvpn2.tesla.com
aws.akamai-apigateway-charging-ownership.tesla.com
digitalassets.tesla.com
inference-staging.bottlerocket.tesla.com
taffy.bottlerocket.tesla.com
fleetview.prd.na.fn.tesla.com
autodiscover.epuca-prd.euw1.vn.cloud.tesla.com
cloud.dpuas-prd.usw.vn.cloud.tesla.com
employeefeedback.tesla.com
forum.analytics-relay.tesla.com
gpv.tesla.com
hermes-stream-api.prd.eu.vn.cloud.tesla.com
edge.digitalassets.tesla.com
citiapipgpprodv7.tesla.com
account.device-api.prd.na.vn.cloud.tesla.com
marketing.tesla.com
status.hermes-stream-api.prd.na.vn.cloud.tesla.com
notebooks.github.tesla.com
pop.akamai-apigateway-suppliers.tesla.com
stream.hermes-api.prd.na.vn.cloud.tesla.com
media.extgithub.tesla.com
files.hermes-stream-api.prd.na.vn.cloud.tesla.com
status.digitalassets.tesla.com
telemetry-ap.prd.vn.cloud.tesla.com
```

subfinder -dL teslasubdomains.txt

Domain list:

```
(dharmik@kali)-[~]
$ subfinder -dL teslasubdomains.txt

subfinder
projectdiscovery.io

[INF] Current subfinder version v2.14.0 (latest)
[INF] Loading provider config from /home/dharmik/.config/subfinder/provider-config.yaml
[INF] Enumerating subdomains for public.apigateway-pricing-gateway.tesla.com
[INF] Found 0 subdomains for public.apigateway-pricing-gateway.tesla.com in 1 second 120 milliseconds
[INF] Enumerating subdomains for gateway-public-service-center-eng5b-fleet.eng.vn.cloud.tesla.com
[INF] Found 0 subdomains for gateway-public-service-center-eng5b-fleet.eng.vn.cloud.tesla.com in 8 seconds 43 milliseconds
[INF] Enumerating subdomains for docker.github.tesla.com
[INF] Found 0 subdomains for docker.github.tesla.com in 21 seconds 42 milliseconds
[INF] Enumerating subdomains for auth-global.tesla.com
[INF] Found 0 subdomains for auth-global.tesla.com in 21 seconds 35 milliseconds
[INF] Enumerating subdomains for dev.device-api.prd.na.vn.cloud.tesla.com
[INF] Found 0 subdomains for dev.device-api.prd.na.vn.cloud.tesla.com in 21 seconds 33 milliseconds
[INF] Enumerating subdomains for digitalassets-contents.tesla.com
[INF] Found 0 subdomains for digitalassets-contents.tesla.com in 21 seconds 31 milliseconds
[INF] Enumerating subdomains for prd.na.vn.cloud.tesla.com
v2.hermes-stream-api.prd.na.vn.cloud.tesla.com
login.device-api.prd.na.vn.cloud.tesla.com
ns2.assistant-api.prd.na.vn.cloud.tesla.com
ns3.device-api.prd.na.vn.cloud.tesla.com
wiki.hermes-api.prd.na.vn.cloud.tesla.com
device-api.prd.na.vn.cloud.tesla.com
gateway.hermes-stream-api.prd.na.vn.cloud.tesla.com
k8s.hermes-stream-api.prd.na.vn.cloud.tesla.com
mail2.hermes-stream-api.prd.na.vn.cloud.tesla.com
stream.hermes-api.prd.na.vn.cloud.tesla.com
vpn.assistant-api.prd.na.vn.cloud.tesla.com
chat.hermes-stream-api.prd.na.vn.cloud.tesla.com
dns1.assistant-api.prd.na.vn.cloud.tesla.com
monitor.hermes-stream-api.prd.na.vn.cloud.tesla.com
my.assistant-api.prd.na.vn.cloud.tesla.com
my.hermes-api.prd.na.vn.cloud.tesla.com
```

httpx

Overview

HTTP probing tool to identify live web services and technologies.

Use Case, Input & Output

Use Case	Filtering a large list of discovered subdomains/hosts down to the ones actually running live web services, and gathering quick metadata on each before deeper manual review.
Input	A list of URLs or hostnames, typically supplied via a file (e.g. subdomains.txt) or piped in through stdin from a tool like Subfinder.
Output	A list of live/responding URLs, optionally enriched with HTTP status codes, page titles, and detected technologies, printed to screen or saved as text/JSON.

Key Features

- HTTP/HTTPS probing
- Status codes
- Titles
- Technology detection

Installation (Kali Linux)

```
sudo apt update
sudo apt install httpx-toolkit -y
httpx -version
```

Status Code	Meaning	Description
200	OK	Website is accessible and working normally.
301	Moved Permanently	Redirects to another URL.
302	Found	Temporary redirect.
403	Forbidden	Server received the request but access is denied.
404	Not Found	The requested page or resource does not exist.
500	Internal Server Error	The server encountered an unexpected error.
503	Service Unavailable	The server is temporarily unavailable or overloaded.

Common Commands

Probe list:

httpx -l teslasubdomains.txt

```
(dharmik@kali)-[~]
$ httpx -l teslasubdomains.txt

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[INF] Current httpx version v1.10.0 (latest)
[WRN] UI Dashboard is disabled, Use -dashboard option to enable
https://analytics-relay.tesla.com
https://akamai-apigateway-vehicleextinfo-gw-prdsvc-st.tesla.com
https://akamai-apigateway-charging-ownership.tesla.com
https://ams13-gpgw1.tesla.com
https://akamai-apigateway-vfx.tesla.com
https://assets-ir.tesla.com
https://akamai-apigateway-suppliers.tesla.com
https://accounts.tesla.com
https://auth.tesla.com
https://apigateway-logging.tesla.com
https://apigateway-message-center-ownership.tesla.com
https://apigateway-charging-bff.tesla.com
https://auth-global.tesla.com
https://apigateway-pricing-gateway.tesla.com
https://auth.prd.euw1.vn.cloud.tesla.com
https://autobidder.powerhub.energy.tesla.com
https://autobidder-preprd.powerhub.energy.tesla.com
https://billing.tesla.com
https://apmv3-cdn-go.tesla.com
https://cdn-design.tesla.com
https://bettertime.tesla.com
https://bettertime-stage.tesla.com
https://charging-transparency.tesla.com
https://auth.eng.usw.vn.cloud.tesla.com
https://auth-stage.tesla.com
https://auth.prd.usw.vn.cloud.tesla.com
https://cua-test-drive-ui.tesla.com
```

```
httpx -l teslasubdomains.txt -status-code
```

Status code:

```
(dharmik@kali)~$ httpx -l teslasubdomains.txt -status-code
[INF] Current httpx version v1.10.0 (latest)
[WRN] UI Dashboard is disabled, Use -dashboard option to enable
https://analytics-relay.tesla.com [403]
https://akamai-apigateway-charging-ownership.tesla.com [503]
https://akamai-apigateway-vehicleextinfo-gw-prdsvc-st.tesla.com [403]
https://akamai-apigateway-vfx.tesla.com [503]
https://akamai-apigateway-suppliers.tesla.com [503]
https://apigateway-logging.tesla.com [503]
https://assets-ir.tesla.com [403]
https://ams13-gpgw1.tesla.com [302]
https://accounts.tesla.com [308]
https://auth-global.tesla.com [403]
https://apigateway-pricing-gateway.tesla.com [503]
https://apigateway-message-center-ownership.tesla.com [401]
https://auth.tesla.com [200]
https://auth.prd.euw1.vn.cloud.tesla.com [200]
https://billing.tesla.com [403]
https://autobidder-preprd.powerhub.energy.tesla.com [403]
https://autobidder.powerhub.energy.tesla.com [403]
https://auth-stage.tesla.com [403]
https://cdn-design.tesla.com [403]
https://bettertime-stage.tesla.com [403]
https://bettertime.tesla.com [403]
https://charging-transparency.tesla.com [403]
https://auth.prd.usw.vn.cloud.tesla.com [200]
https://auth.eng.usw.vn.cloud.tesla.com [200]
https://courses.tesla.com [403]
https://apigateway-charging-bff.tesla.com [401]
https://cua-help-me-charge-ui.tesla.com [403]
```

httpx -l subdomains.txt -title

Title:

```
(dharmik@kali)-[~]
$ httpx -l teslasubdomains.txt -title

[INF] Current httpx version v1.10.0 (latest)
[WRN] UI Dashboard is disabled, Use -dashboard option to enable
https://akamai-apigateway-vehicleextinfo-gw-prdsvc-st.tesla.com [Access Denied]
https://accounts.tesla.com [Access Denied]
https://analytics-relay.tesla.com [Access Denied]
https://akamai-apigateway-charging-ownership.tesla.com [Service Unavailable]
https://akamai-apigateway-vfx.tesla.com [Service Unavailable]
https://akamai-apigateway-suppliers.tesla.com [Service Unavailable]
https://ams13-gpgw1.tesla.com
https://apigateway-logging.tesla.com [Service Unavailable]
https://assets-ir.tesla.com [Access Denied]
https://auth-stage.tesla.com [Access Denied]
https://autobidder-preprd.powerhub.energy.tesla.com [Access Denied]
https://auth.tesla.com [Access Denied]
https://auth-global.tesla.com [Access Denied]
https://auth.prduw1.vn.cloud.tesla.com [Tesla Auth]
https://apigateway-pricing-gateway.tesla.com [Service Unavailable]
https://apigateway-charging-bff.tesla.com
https://apigateway-message-center-ownership.tesla.com
https://bettertime.tesla.com [Access Denied]
https://billing.tesla.com [Access Denied]
https://cdn-design.tesla.com [Access Denied]
https://autobidder.powerhub.energy.tesla.com [Access Denied]
https://charging-transparency.tesla.com [Access Denied]
https://bettertime-stage.tesla.com [Access Denied]
https://auth.prduw.vn.cloud.tesla.com [Tesla Auth]
https://cua-chat-ui.tesla.com [Access Denied]
https://courses.tesla.com [Access Denied]
https://auth.eng.usw.vn.cloud.tesla.com [Tesla Auth]
```

httpx -l subdomains.txt -tech-detect

Tech detect:

```
(dharmik@kali)-[~]
$ httpx -l teslasubdomains.txt -tech-detect

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[INF] Current httpx version v1.10.0 (latest)
[WRN] UI Dashboard is disabled, Use -dashboard option to enable
https://analytics-relay.tesla.com [HSTS]
https://akamai-apigateway-vehicleextinfo-gw-prdsvc-st.tesla.com [HSTS]
https://akamai-apigateway-vfx.tesla.com [HSTS]
https://ams13-gpgw1.tesla.com [HSTS]
https://akamai-apigateway-charging-ownership.tesla.com [HSTS]
https://accounts.tesla.com [Akamai,Akamai Bot Manager,HSTS,Varnish]
https://apigateway-logging.tesla.com [HSTS]
https://auth-stage.tesla.com
https://apigateway-pricing-gateway.tesla.com [HSTS]
https://autobidder-preprd.powerhub.energy.tesla.com [HSTS]
https://assets-ir.tesla.com [HSTS]
https://apigateway-charging-bff.tesla.com [Akamai,HSTS]
https://akamai-apigateway-suppliers.tesla.com [HSTS]
https://apigateway-message-center-ownership.tesla.com [Akamai,HSTS]
https://auth-global.tesla.com [Akamai,Akamai Bot Manager]
https://bettertime-stage.tesla.com [HSTS]
https://bettertime.tesla.com [HSTS]
https://autobidder.powerhub.energy.tesla.com [HSTS]
https://auth.prd.euw1.vn.cloud.tesla.com [C3.js,Envoy,HSTS,hCaptcha]
https://billing.tesla.com [HSTS]
https://cdn-design.tesla.com [HSTS]
https://charging-transparency.tesla.com [HSTS]
https://auth.tesla.com [Akamai,Akamai Bot Manager,C3.js,Envoy,HSTS,hCaptcha]
https://auth.prd.usw.vn.cloud.tesla.com [C3.js,Envoy,HSTS,hCaptcha]
https://auth.eng.usw.vn.cloud.tesla.com [Envoy,HSTS,hCaptcha]
https://cua-chat-ui.tesla.com [HSTS]
https://cua-help-me-charge-ui.tesla.com [HSTS]
```

Amass

Overview

OWASP attack surface mapping and advanced subdomain discovery framework.

Use Case, Input & Output

Use Case	In-depth attack surface mapping that combines passive OSINT with active DNS and network probing to build a thorough picture of an organization's external footprint.
Input	A target domain name; optional configuration file with API keys, resolvers, and wordlists for brute-force enumeration.
Output	Discovered subdomains, associated IP addresses, and ASN/network ownership data, displayed on screen or written to an output file for further analysis.

Key Features

- Passive & active enumeration
- DNS intelligence
- ASN mapping
- Attack surface mapping

Installation (Kali Linux)

```
sudo apt update
sudo apt install amass -y
amass version
```

Common Commands

Passive:	amass enum -passive -d example.com 
Active:	amass enum -active -d example.com 
Bruteforce:	amass enum -brute -d example.com 
Save:	amass enum -d example.com -o result.txt

Recommended Reconnaissance Workflow

1. Recon-ng — Collect OSINT
2. Subfinder — Enumerate subdomains
3. Amass — Expand attack surface
4. httpx — Identify live web services
5. Nmap — Scan ports and services
6. Vulnerability assessment and reporting